AI Playbook and Toolkit for Public Health Departments

Retrieval-Augmented Generation for Public Health Knowledge Work

ARC 300

Retrieval-augmented generation, often called RAG, combines generative AI with retrieval from a defined set of documents or data sources. Instead of relying only on a model's general training, a RAG system searches approved content and uses retrieved material to generate an answer or summary. In public health, RAG may support policy lookup, protocol interpretation, internal help desks, training support, grant review, emergency response, or program knowledge management. RAG can make generative AI more useful and safer when it is designed well. It can ground responses in agency documents, cite sources, and reduce the risk of unsupported answers. However, RAG does not automatically solve hallucination, privacy, access control, outdated guidance, conflicting documents, or poor retrieval. The system is only as reliable as the content collection, search process, permissions, and review workflow. This module teaches learners the core components of RAG and how to evaluate whether a RAG tool is ready for public health use.

Level	intermediate/practitioner
Primary track	Technical Architecture Track
Appears in tracks	analytics-modeling

Learning Objectives

- Define retrieval-augmented generation in plain language.
- Describe the major components of a RAG system, including document ingestion, chunking, embeddings, vector search, retrieval, generation, and citation.
- Identify public health use cases where RAG may improve knowledge work.
- Assess risks related to stale content, access control, retrieval quality, hallucination, and source misuse.
- Design governance and monitoring requirements for agency RAG systems.
- Produce a RAG readiness checklist for a public health knowledge workflow.

Module Overview

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Why This Topic Matters for Public Health AI

RAG matters because public health agencies manage large collections of guidance, protocols, policies, contracts, plans, reports, forms, FAQs, and training materials. Staff may spend substantial time searching for answers across shared drives, websites, portals, and outdated files. A well-designed RAG system can help users find relevant guidance and produce concise, source-linked answers.

The public health value of RAG depends on governance. If a RAG tool includes outdated guidance, draft policies, documents outside a user's permission level, or conflicting versions, it may generate misleading answers. If it fails to retrieve the most relevant source, the generated answer may be incomplete or wrong. RAG improves the grounding problem only when the content collection and retrieval process are managed carefully.

RAG is especially important for responsible AI because it offers a middle path between generic public AI tools and fully custom systems. Agencies can build assistants around approved content, but they must define content ownership, update schedules, access rules, citation requirements, testing, monitoring, and user responsibilities.

Public Health Example

A state health department may create an internal RAG assistant to help staff find current infectious disease investigation protocols. The assistant could search approved protocols, case definitions, reporting rules, and escalation procedures. When a user asks what steps to take for a suspected condition, the system could summarize relevant guidance and cite the source documents.

This workflow requires strong controls. The assistant must include only current approved guidance, respect role-based access, cite sources, and warn users when content is outdated or missing. Staff should verify the cited protocol before acting, especially when the situation is unusual or high risk.

Technical Deep Dive

A RAG system begins with content selection. Agencies should decide which documents are authoritative, current, approved for the use case, and appropriate for the intended users. Content inventories should include owner, version, effective date, review date, sensitivity level, and access rules. Without this foundation, retrieval can surface obsolete or unauthorized material.

Document processing affects retrieval quality. Files may need to be converted, cleaned, segmented, and indexed. Chunking decisions matter because overly small chunks may omit context while overly large chunks may retrieve irrelevant material. Tables, figures, scanned PDFs, forms, and appendices may require special handling. Public health documents often contain complex context, such as exceptions, legal caveats, and jurisdiction-specific rules.

Embeddings and vector search help identify semantically similar content, but retrieval should be tested. Staff should use realistic questions and evaluate whether the system retrieves the best sources. Testing should include synonyms, acronyms, misspellings, role-specific questions, edge cases, and questions that should produce “not found” or escalation responses.

Generation should be constrained by retrieved material. The system should be instructed to answer from sources, cite them, identify uncertainty, and avoid unsupported claims. It should also explain when guidance conflicts or when source material is insufficient. Users should be able to open the cited source and verify key statements.

Monitoring should include retrieval failures, unsupported answers, stale content, user feedback, access-control incidents, and frequently asked questions. Content governance should define how documents are added, removed, updated, archived, and re-indexed.

Risks, Failure Modes, and Guardrails

Stale or conflicting content. Outdated or inconsistent documents can produce wrong answers. Guardrails include content ownership, version control, review dates, and archival rules.

Permission leakage. RAG may retrieve content a user should not see. Guardrails include role-based access, document-level permissions, and audit logs.

Poor retrieval. The system may retrieve irrelevant or incomplete chunks. Guardrails include test question sets, retrieval evaluation, chunking review, and user feedback.

Unsupported generation. The model may add claims beyond retrieved sources. Guardrails include source-only instructions, citation requirements, and reviewer verification.

Application to AI-Supported Workflows

RAG supports generative AI by grounding outputs in defined sources. It may be used in chat assistants, training tools, policy support, emergency operations, procurement review, and program knowledge bases. It can also support agentic workflows by giving agents access to approved guidance, but actions based on retrieved content require additional guardrails.

Public health RAG systems should be built around trust. Users should know what content is included, what is excluded, how current it is, whether they have permission to use it, and when they must verify or escalate.

Reflection Questions

What content collection would the RAG system use?

Who owns and updates each source document?

How will access permissions be enforced?

How will retrieval quality and citation accuracy be tested?

What should the system do when sources are missing, outdated, or conflicting?

Practical Exercise

Create a RAG readiness checklist for one public health knowledge workflow. Include content inventory, source authority, document owners, update schedule, sensitivity level, access controls, chunking concerns, retrieval test questions, citation requirements, user review expectations, and monitoring indicators.

The checklist should identify at least three “do not answer” or escalation situations, such as missing guidance, conflicting sources, or questions outside approved scope.

Expected Artifact or Evidence

RAG readiness checklist

Content inventory and ownership table

Retrieval test question set

Citation and source verification protocol

Expected Assignment

- RAG readiness checklist
- Content inventory and ownership table
- Retrieval test question set
- Citation and source verification protocol

Knowledge Check

- 1. What is RAG?
- 2. Why does chunking matter?
- 3. What is a major RAG governance risk?
- 4. What should a RAG system do when sources are insufficient?
- 5. What is strong completion evidence for this module?

References and Resources

- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- OWASP Top 10 for Large Language Model Applications: https://owasp.org/www-project-top-10-for-large-language-model-applications/
- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- ONC USCDI: https://isp.healthit.gov/united-states-core-data-interoperability-uscdi